Who Runs All Those Containers?

Kubernetes: orchestrating containers

SCAN / LEARN

THE MAIN IDEA

Kubernetes is software that coordinates containers across computers. It observes the desired configuration and tries to keep actual running services close to it.

FOLLOW THE SYSTEM

01 DESIRED STATE

A service should have three running copies.

02 SCHEDULER

Kubernetes places work on suitable machines.

03 OBSERVE

It checks which copies are healthy.

04 RECONCILE

It restarts or replaces missing copies.

DESIRED STATE → PLACE → CHECK → REPAIR

COMMON MISCONCEPTION

Kubernetes guarantees that every application works.

MORE ACCURATE

It manages deployment and recovery, not faulty business logic.

THREE QUESTIONS TO KEEP

PROBLEM: How do we coordinate many containers?

FAILURE: Bad settings may restart the wrong things.

IN REAL LIFE: Cloud clusters and large applications.

SOLO DISCOVERY

Draw 3 machines and place 5 tasks with limited space.